

Mathilde Gerbelli-Gauthier

Curriculum Vitae

Department of Mathematics
Bahen Centre for Information Technology, University of Toronto
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Positions Held

- 2024- **Assistant Professor**, *University of Toronto*
- 2020-2024 **Postdoctoral Fellow**, *Centre de Recherches Mathématiques and McGill University*
- 2021-2022 **Member**, *Institute for Advanced Study*

Education

- 2020 **Ph.D. in Mathematics**, *University of Chicago*, Advisor: Matthew Emerton
Thesis: Growth of Cohomology of Arithmetic Groups and the Stable Trace Formula.
- 2014 **B.Sc. in Mathematics**, *McGill University*, First Class Honours

Grants and Awards

- 2026 **Connaught New Researcher Award**, University of Toronto, 30,000 CAD
- 2025 **Discovery Grant**, NSERC, 185,000 CAD
Discovery Launch Supplement, NSERC, 12,500 CAD
- 2020 **Postdoctoral Fellowship**, NSERC
Kowalsky Fellowship, Department of Mathematics, University of Chicago
- 2018 **Lawrence and Josephine Graves Prize for Excellence in Undergraduate Teaching**,
Department of Mathematics, University of Chicago
- 2016 **Postgraduate Scholarship - Doctoral**, NSERC
Alexander Graham Bell Canada Graduate Scholarship, NSERC, *declined*

Publications

- 2026 **Mod ℓ gamma factors and a converse theorem for finite general linear groups**, with J. Bakeberg, H. Goodson, A. Iyengar, G. Moss, and R. Zhang
Documenta Mathematica, **31**:1
- 2025 **Large sums of Fourier coefficients of cusp forms**, with C. Frechette, A. Hamieh, and N. Tanabe
Journal de Théorie des Nombres de Bordeaux, **37**:1
Cohomology of Fuchsian groups and Fourier interpolation, with A. Venkatesh
Essential Number Theory, **4**:2
- 2023 **Limit multiplicity for unitary groups and the stable trace formula**
Algebra and Number Theory, **17**:12
- 2016 **The order of nilpotence of Hecke operators mod 2: a new proof**
Research in Number Theory, **2**:7

Preprints

- 2025 **Uniform bounds and uncertainty for asymptotics of representations of p -adic GL_N** ,
with R. Dalal and S. Marshall
arXiv:2511.11063
- 2024 **A note on large sums of divisor-bounded multiplicative functions**, with C. Frechette,
A. Hamieh, and N. Tanabe
To appear in *Research Directions in Number Theory*. ArXiv: 2405.00658
- Root Number Equidistribution for Self-Dual Automorphic Representations on GL_N** ,
with R. Dalal
Submitted. ArXiv: 2410.01976
- 2023 **The Cohomological Sarnak-Xue Density Hypothesis for SO_5** , with S. Evra and H.
Gustafsson
Submitted. ArXiv: 2309.12413
- 2022 **Statistics of cohomological automorphic representations via the endoscopic classification**, with R. Dalal
Submitted. ArXiv: 2212.12138

Talks

- 2026 The Australian Direction: A conference honoring the mathematical contributions of Matt Emerton and Mark Kisin, University of Utah.
Séminaire Groupes Réductifs et Formes Automorphes, Institut Mathématique de Jussieu - Paris Rive Gauche.
Group Actions and Harmonic Analysis in Number Theory, Oberwolfach Research Institute for Mathematics.
Algebra Seminar, IMPA.
Departmental Colloquium, University of Minnesota.
AMS Spring Sectional Meeting: Special Session on Automorphic Representations, Boston College.
AMS Spring Sectional Meeting: Special Session on Shimura Varieties, Boston College.
Number Theory Seminar, University of Waterloo.
Departmental Colloquium, Queen's University.
- 2025 Trace Formulas: 7th Nisyros Conference on Automorphic Forms and Related Topics, Nisyros.
CMS Summer Meeting, Université Laval.
Joint Columbia-CUNY-NYU Number Theory Seminar, CUNY Graduate Center.
Sphere-packing and beyond: celebrating Maryna Viazovska's honoris causa at Sorbonne Université, Sorbonne Université. *Remote*
Departmental Colloquium, University of Illinois at Chicago.
Algebra Seminar, Michigan State University.
Departmental Colloquium, University of Toronto.
- 2024 GANITA Seminar, University of Toronto.
Number Theory Seminar, University of Toronto.
Québec-Maine Number Theory Conference, Université Laval.
Canadian Number Theory Association XVI, Fields Institute.
Women⁺ and Mathematics Program: Postdoc Talks, Institute for Advanced Study.
Bhaskaracharya celebrating Harish-Chandra, Bhaskaracharya Pratishthana and IISER Pune, *Online*.
Ottawa-Carleton Number Theory Seminar, University of Ottawa.

- Beyond Voganish Seminar, University of Calgary, *Online*.
- AMS Spring Sectional: Special Session on Automorphic Forms and Trace Formulae, Howard University.
- QVNTS, McGill University.
- Number Theory Seminar, Rutgers University, *Online*.
- Colloquium, Boston College.
- Undergraduate Colloquium, Bowdoin College.
- Colloquium, University of Utah.
- Representation Theory/Number Theory Seminar, University of Utah.
- 2023 Colloquium, University of Toronto.
- Colloquium, Georgia Tech.
- Number Theory Seminar, Johns Hopkins University.
- Maine-Québec Number Theory Conference, University of Maine.
- Heilbronn Number Theory Seminar, University of Bristol.
- Clay Workshop: New Perspectives in the Analytic Theory of Automorphic Forms, Oxford.
- Conference on Automorphic forms, Endoscopy and Trace Formulas, CIRM.
- International Automorphic Forms Seminar, *Online*.
- Group, Lie, and Number Theory Seminar, University of Michigan.
- Number Theory Seminar, Northwestern University.
- Number Theory Seminar, University of Chicago.
- AMS Spring Sectional Meeting: Special Session on Arithmetic Statistics, University of Cincinnati.
- Number Theory Speaker Series, Ohio State University.
- Automorphic Forms and Arithmetic Seminar, Columbia University.
- Princeton-IAS Number Theory Seminar, Princeton University.
- COGENT Seminar, *Online*.
- Joint Mathematics Meeting: Automorphic Forms and Representation Theory, Boston.
- 2022 Modern Math Workshop, San Juan.
- Québec-Maine Number Theory Conference, Université Laval.
- Connecticut Summer School in Number Theory, University of Connecticut.
- "What is..." Seminar, Institute for Advanced Study.
- Philadelphia Area Number Theory Seminar, Bryn Mawr College.
- Mathematics Colloquium, Dalhousie University. *Remote*
- 2021 Automorphic Forms and Number Theory Seminar, Purdue University. *Remote*
- Members' Colloquium, Institute for Advanced Study.
- Workshop on Cohomology of Arithmetic Groups: Duality, Stability, and Computations, Banff.
- Boston University Number Theory Seminar. *Remote*
- Séminaire représentations des groupes réductifs, Institut Mathématique de Marseille. *Remote*
- CMS 75th+1 Anniversary Summer Meeting, Ottawa. *Remote*
- Séminaire de géométrie arithmétique et motivique, Université Sorbonne-Paris-Nord. *Remote*
- Arbeitsgemeinschaft: Derived Galois Deformation Rings and Cohomology of Arithmetic Groups, Oberwolfach. *Remote*.
- Joint IAS/Princeton Number Theory Seminar, *Remote*.
- (w/ Cay del Junco) CONSTRAIN Researcher Forum, University of Leeds. *Remote*.

- 2020 CMS Virtual Winter Meeting, University of Ottawa. *Remote*.
 Number Theory Seminar, University of Oregon. *Remote*.
 Québec-Vermont Number Theory Seminar, McGill University. *Remote*.
 Members' Seminar, CRM Special Semester on Cohomology in Arithmetic. *Remote*
 Number Theory Seminar, University of Wisconsin-Madison.
 Representation Theory and Number Theory Seminar, University of Utah.
 Group, Lie, and Number Theory Seminar, University of Michigan.
- 2019 Junior Number Theory Days, Johns Hopkins University.
 Number Theory Seminar, MIT.
 Number Theory Seminar, Johns Hopkins University.
 AMS Fall Sectional Meeting: Special Session on Automorphic Forms and L -functions, University of Wisconsin–Madison.
 Linfoot Seminar, University of Bristol.
- 2018 Women in Automorphic Forms, Technische Universität Darmstadt.

Selected Workshops and Conferences

- 2026 **The Australian Direction: A conference honoring the mathematical contributions of Matt Emerton and Mark Kisin**, *Salt Lake City*
Group Actions and Harmonic Analysis in Number Theory, *Oberwolfach*
- 2025 **7th Nisyros Conference on Automorphic Forms and Related Topics**, *Nisyros*
Sphere-packing and beyond: celebrating Maryna Viazovska's honoris causa at Sorbonne Université, *Paris*, *Remote*
- 2024 **Québec-Maine Number Theory Conference**, *Québec*
Women⁺ and Mathematics: Symmetry and Arithmetic, *Princeton*
- 2023 **Clay Workshop: New Perspectives in the Analytic Theory of Automorphic Forms**, *Oxford*
Conference on Automorphic forms, Endoscopy and Trace Formulas, *Marseille*
Women in Numbers 6, *Banff*
Joint Mathematics Meeting, *Boston*
- 2022 **Modern Math Workshop and National Diversity in STEM Conference**, *San Juan*
Québec-Maine Number Theory Conference, *Québec*
Connecticut Summer School in Number Theory, *Storrs*
Arizona Winter School: Automorphic Forms Beyond GL_2 , *Tucson*
- 2021 **Cohomology of Arithmetic Groups: Duality, Stability, and Computations**, *Banff*
Rethinking Number Theory, *Online*
CMS 75th+1 Anniversary Summer Meeting, *Ottawa*, *Remote*
Arbeitsgemeinschaft: Derived Galois Deformation Rings and Cohomology of Arithmetic Groups, *Oberwolfach*, *Remote*
- 2020 **CMS Winter Meeting**, *Ottawa*, *Remote*
CRM Workshop: Arithmetic Quotients of Symmetric Spaces and Their Cohomology, *Montreal*
- 2019 **CMO Workshop: Modularity and Moduli Spaces**, *Oaxaca*
A Conversation on Professional Norms in Mathematics, *Baltimore*

- AMS Fall Sectional Meeting: Special session on Automorphic Forms and L -functions**, *Madison*
- 2018 **Sixth Abel Conference: A Mathematical Celebration of Robert P. Langlands**, *Minneapolis*
Women in Automorphic Forms, *Darmstadt*
IAS Workshop: Representation Theory and Analysis on Locally Symmetric Spaces, *Princeton*
- 2017 **IAS Workshop: Motives, Galois Representations and Cohomology Around the Langlands Program**, *Princeton*
MSRI Summer School: Automorphic Forms the Langlands Programme, *Berkeley*
Arizona Winter School: Perfectoid Spaces, *Tucson*
Barcelona Graduate School of Mathematics: p -adic Methods in Galois Representations and Modular Forms, *Barcelona*

Teaching

Instructor

University of Toronto

Algebra I, Topics in Algebra: Representation Theory of Reductive p -adic Groups, Aha! Mathematical Discovery and Creative Problem Solving.

McGill University

Linear Algebra and Geometry.

University of Chicago

Mathematical Methods for the Social Sciences, Calculus III.

Workshops Attended

- 2023 Equity Discussion Group, *Department of Mathematics and Statistics, McGill University*.
 2022 Pedagogy Discussion Group, *Department of Mathematics and Statistics, McGill University*.
 2019 Fundamentals of Teaching Mathematics, *Chicago Center for Teaching*.
 2018 Inclusive Teaching in STEM, *Race and Pedagogy working group at the University of Chicago*.

Supervision

Postdoctoral Fellows

- 2025- Yi Shan

Undergraduate Students

- 2025 Ceren Sahin
 2025 Liam Gauvreau

Service and Outreach

Refereeing

I have served as a referee for Transactions of the American Mathematical Society, GAFA, Essential Number Theory, and Commentarii Mathematici Helvetici.

Volumes Edited

- 2021 **A Conversation on Professional Norms in Mathematics**, *American Mathematical Society*, ISBN: 978-1-4704-6780-7
Burnout: a zine about academia, travel, and climate change, *self-published*, knowledge.uchicago.edu/record/2773

Conferences and Seminars Organized

- 2023 **Scientific Session: Automorphic representations and p -adic aspects of the Langlands program**, *CMS Winter Meeting*, Co-organizer.
AMS Special Session: Women in Automorphic Forms, *Joint Mathematics Meeting*, Co-organizer.
- 2022 **Mathematical Conversations**, *Institute for Advanced Study*, Co-organizer
- 2020 **CRM Women's Seminar**, *Centre de Recherche Mathématiques*, Co-organizer
- 2016-2017 **Student Number Theory Seminar**, *University of Chicago*, Organizer
Fall-Winter 2016: p -adic geometry, Summer 2017: the Local Langlands correspondence for GL_2

Other

- 2025 **OURFAM: Online Undergraduate Resource Fair for the Advancement and Alliance of Marginalized Mathematician**, *Online*, *Our Stories* speaker
Presented elements of my personal/mathematical journey
- 2024 **Secrets de surface: discussion**, *Coeur des sciences de l'UQAM*, Panelist
Part of a discussion panel on Maryam Mirzakhani's legacy and women in Mathematics.
- 2023 **EDI assesment**, *Banff International Research Station*, Evaluator
Assess the merits of workshop proposals on the basis of Equity, Diversity, and Inclusion.
- 2022-2023 **Part B Exam**, *McGill University*, Secondary Examiner
Topics: Growth in the torsion of cohomology of arithmetic groups, Hilbert modular forms.
- 2022 **Lunch in the Time of Covid**, Panelist
Discussion on the role and responsibility of academics in climate change.
- 2021-2022 **Women's Lunches**, *Institute for Advanced Study*, Co-organizer
Organized a monthly lunch meeting for female members at the Institute's School of Mathematics.
- 2019-2020 **Arts + Science Collaboration Grant**, *University of Chicago*, Fellow
Recipient of a \$3000 grant for an interdisciplinary arts and science project with two students at the School of the Art Institute of Chicago. Title: *Numerical Alternatives*
- 2018-2020 **Lunchtime Ado with Female Speakers**, *University of Chicago*, Co-organizer
Organized and moderated a series of biquarterly lunchtime conversations with women speaking in the department's research seminars. Recipient of an \$800 grant in 2019-2020.
- 2017-2020 **Directed Reading Program**, *University of Chicago*, Co-organizer
Managed a program pairing undergraduate and graduate students for quarter-long reading projects.
- 2018-2019 **Graduate Students United**, *University of Chicago*, Steward
Union representative for graduate workers in the mathematics department.
- 2018 **Department of Mathematics Climate Survey**, *University of Chicago*
Co-development of a climate survey to assess the experiences of graduate students.
- 2016-2018 **Directed Reading Program**, *University of Chicago*, Mentor
Projects: Class field theory, Pythagorean triples, Fermat's theorem on the sum of two squares, Algebraic Varieties, Fourier Series, Arithmetic of Elliptic Curves.
- 2013-2014 **Program in Mathematics for Young Scientists**, *Boston University*, Counselor
Mentorship of high-school students in a 6-week number theory summer camp.